

Lecture 21.

Eigenvalues - Eigenvectors.

Ax parallel to x

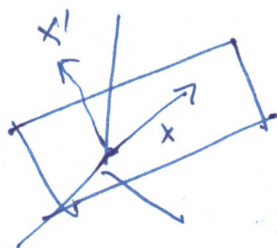
$$Ax = \lambda x$$

If A is singular, $\lambda = 0$ is e-value.

⊥ What are x 's and λ 's for projection matrix P

① Any x in plane: $Px = x$ ($\lambda = 1$)

② Any $x \perp$ plane: $Px = 0$ ($\lambda = 0$)



Permutation

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \lambda = 1, Ax = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = x.$$

$$x = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \lambda = -1, Ax = \begin{bmatrix} -1 \\ 1 \end{bmatrix} = -x$$

Fact: Sum of λ 's =



$a_{11} + a_{22} + \dots + a_{nn}$ (Diagonal values). (Trace)

prod of λ 's = $\det A$

How to solve $Ax = \lambda x$

Rewrite: $(A - \lambda I)x = 0$

↑ singular,

$$\det(A - \lambda I) = 0$$

Find λ first.

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \quad \det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix}$$

$$= (3-\lambda)^2 - 1$$

$$\text{Trace} = 6 \quad = \lambda^2 - 6\lambda + 8 = (\lambda-2)(\lambda-4)$$

$$\lambda_1 = 4 \quad \lambda_2 = 2 \quad \delta \text{ is } \det(A)$$

$$A - 4I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$A - 2I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

If $Ax = \lambda x$
then $(A + \beta I)x = (\lambda + \beta)x$

If $Ax = \lambda x$, $By = \alpha y$

$$\cancel{(A+B)x = (\lambda+\alpha)x}$$

because eigenvectors
are usually different

Example $Q = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ trace $= 0+0 = \lambda_1 + \lambda_2$
det $= 1 = \lambda_1 \cdot \lambda_2$

90° rotation

$$\det(Q - \lambda I) = \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 + 1 = 0$$

$$\lambda_1 = i$$

$$\lambda_2 = -i$$

symmetric $n \times n$
have real eigenvalue.

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix} \quad \det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 1 \\ 0 & 3-\lambda \end{vmatrix} = (3-\lambda)(3-\lambda)$$

$$(A - \lambda I)x = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\lambda_1 = 3$$

$$\lambda_2 = 3$$

$$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

No
 2^n independent eigen vectors.

Lecture 22.
Diagonalize A $S^{-1}AS = \Lambda$

$A - \lambda I$ singular

$$Ax = \lambda x$$

Suppose n eigen vectors of A (Independent)
put them in columns of S

diagonal eigenvalue matrix.

$$AS = A \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix} = \begin{bmatrix} \lambda_1 x_1 & \dots & \lambda_n x_n \end{bmatrix} = \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} = S \Lambda$$

Only if S independent!!

$$AS = S\Lambda$$

$$S^{-1}AS = \Lambda$$

$$\rightarrow A = S\Lambda S^{-1}$$

$$\text{If } Ax = \lambda x$$

$$A^2 x = \lambda Ax = \lambda^2 x \leftarrow A^2 \text{'s eigen value} = \lambda^2$$

eigen vector = x

$$A^2 = S\Lambda S^{-1}S\Lambda S^{-1} = S\Lambda^2 S^{-1} \quad \text{same interpretation}$$

$$\boxed{A^k = S\Lambda^k S^{-1}}$$

★ Eigen value give great insight of matrix's power!

Theorem $A^k \rightarrow 0$ as $k \rightarrow \infty$ if all eigen value $|\lambda_i| < 1$

A is sure to have n independent e vectors
(and be diagonalizable)
if all the λ 's are different
(i.e. no repeated λ s)

Repeated eigenvalues // may or may not have n indep e vectors
i.e. I.

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \quad \det \begin{bmatrix} 2-\lambda & 1 \\ 0 & 2-\lambda \end{bmatrix} = 0$$

$$\lambda = 2, 2.$$

$$A - 2I = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{null space} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

★ Linear Algebra. 23

Differential Eqns $\frac{du}{dt} = Au$

$$u(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Example:

$$\frac{du_1}{dt} = -u_1 + 2u_2$$

$$\frac{du_2}{dt} = u_1 - 2u_2$$

$$\rightarrow A = \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix}$$

$$A = \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix} \quad \lambda_1 = 0 \quad x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad Ax_1 = 0x_1$$

$$\lambda = 0, -3 \quad \lambda_2 = -3 \quad x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad Ax_2 = -3x_2$$

Solutions = $u(t) = C_1 e^{\lambda_1 t} x_1 + C_2 e^{\lambda_2 t} x_2$

Check: $\frac{du}{dt} = Au$, $\lambda_1 e^{\lambda_1 t} x_1 = A e^{\lambda_1 t} x_1$

$$C_1 \lambda_1^k x_1 + C_2 \lambda_2^k x_2 \rightarrow u_{k+1} = A u_k$$

use $u(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = C_1 \cdot 1 \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} + C_2 e^{-3 \cdot 0} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

At $t=0$, $C_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + C_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \leftarrow Sc = U_0$

Solution: $C_1 = \frac{1}{3} \quad C_2 = \frac{1}{3}$

find answer $u(t) = \frac{1}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \frac{1}{3} e^{-3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Steady state
 $u(\infty) = \frac{1}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

① Stability: $u(t) \rightarrow 0$ / need $e^{\lambda t} \rightarrow 0$ / $\text{Re } \lambda < 0$

② Steady state: $\lambda_1 = 0$ and other $\text{Re } \lambda < 0$

③ blow up if and $\text{Re } \lambda > 0$

2x2 stability $\operatorname{Re} \lambda_1 < 0$ $\operatorname{Re} \lambda_2 < 0$

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$: trace $a+d = \lambda_1 + \lambda_2 < 0$, complex part will be conjugative.
 $\det > 0$. ($\lambda_1 \lambda_2 > 0$)

Set $u = Sv$

$$\frac{du}{dt} = Au$$

$$S \frac{dv}{dt} = ASv$$

$$\frac{dv}{dt} = S^{-1}ASv = \Lambda v \quad \left| \quad \begin{array}{l} \frac{dv_1}{dt} = \lambda_1 v_1 \\ \vdots \end{array} \right. \quad v(t) = e^{\Lambda t} v(0)$$

$$u(t) = S e^{\Lambda t} S^{-1} u(0)$$

Matrix exponential $e^{At} = I + At + \frac{(At)^2}{2} + \frac{(At)^3}{6} + \dots + \frac{(At)^n}{n!} + \dots$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$
$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

$$(I - At)^{-1} = I + At + (At)^2 + (At)^3 + \dots$$

Lecture 24. Markov matrixes.

Steady state : $\lambda = 1$

$$A = \begin{bmatrix} .1 & .01 & .3 \\ .2 & .99 & .3 \\ .7 & .0 & .4 \end{bmatrix}$$

① All entries ≥ 0

② All columns add to 1.

1. $\lambda = 1$ is an eigenvalue

2. All other $|\lambda_c| < 1$

$$U_k = A^k U_0 = c_1 \underset{\substack{\uparrow \\ \lambda_1 = 1}}{\lambda_1^k} x_1 + c_2 \underset{\substack{\uparrow \\ |\lambda_2| < 1}}{\lambda_2^k} x_2 + \dots \rightarrow c_1 \underset{\substack{\downarrow \\ \text{Steady state.}}}{x_1} \text{ part of } U_0$$

$$A - I = \begin{bmatrix} -.9 & .01 & .3 \\ .2 & -.01 & .3 \\ .7 & 0 & -.6 \end{bmatrix}$$

$\rightarrow$ columns

because $(1, 1, 1)$ is in $N(A-I)$
 then x_1 is in $N(A-I)$

All columns of $A-I$ add to zero. $\rightarrow A-I$ is singular

(because rows are dependent!)

eigenvalues of A $\det(A - \lambda I) = 0$

eigenvalues of A^T

are the same

$$\Downarrow$$

$$\det(A^T - \lambda I) = 0$$

$U_{k+1} = A U_k$. A is Markov

$$\begin{bmatrix} U_{cal} \\ U_{mass} \end{bmatrix}_{t=k+1} = \begin{bmatrix} .9 & .2 \\ .1 & .8 \end{bmatrix} \begin{bmatrix} U_{cal} \\ U_{mass} \end{bmatrix}_{t=k}$$

$$\begin{bmatrix} U_{cal} \\ U_{mass} \end{bmatrix}_0 = \begin{bmatrix} 0 \\ 1000 \end{bmatrix} \quad \begin{bmatrix} U_{cal} \\ U_{mass} \end{bmatrix}_1 = \begin{bmatrix} 200 \\ 800 \end{bmatrix}$$

$$\begin{bmatrix} .9 & .2 \\ -.1 & .8 \end{bmatrix} \quad \lambda_1 = 1 \quad x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\lambda_2 = 0.7 \quad x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$U_k = C_1 1^k \begin{bmatrix} 2 \\ 1 \end{bmatrix} + C_2 0.7^k \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$U_0 = \begin{bmatrix} 0 \\ 1000 \end{bmatrix} = C_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + C_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$C_1 = \frac{1000}{3}$$

$$C_2 = \frac{2000}{3}$$

Fourier Series / Projections with orthonormal matrix basis
 $\nwarrow$ expansion $q_1 \dots q_n$

$$\text{Any } v = x_1 q_1 + x_2 q_2 + \dots + x_n q_n$$

$$q_i^T v = x_1 q_i^T q_1 + 0 + \dots + 0 = x_1$$

$$\rightarrow \begin{bmatrix} q_1 & \dots & q_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = v$$

$$\text{Q } x = v.$$

$$x = Q^T v = Q^T v$$

$$\boxed{x_i = q_i^T v}$$

Fourier Series: $f(x) = \frac{1}{2} (f(x+2\pi) + f(x-2\pi))$

$$f(x) = a_0 + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + \dots$$

inner product = $\nwarrow$ basis

vectors:

$$v^T w = v_1 w_1 + v_2 w_2 + \dots + v_n w_n$$

functions:

$$f^T g = \int_0^{2\pi} f(x) g(x) dx$$

$$\int_0^{2\pi} \cos x \sin x dx = \frac{1}{2} \sin^2 x \Big|_0^{2\pi} = 0$$

Linear Algebra 25

symmetric matrices:

① eigen values are real

② eigen vectors are orthogonal!!

(can be chosen)

Usual: $A = S \Lambda S^{-1}$
case

orthonormal eigen
vectors

Symmetric case $A = Q \Lambda Q^T = \boxed{Q \Lambda Q^T}$

↑ spectral theorem

Proof: Why Real eigen values?

$$Ax = \lambda x \xrightarrow{\text{always}} \bar{A} \bar{x} = \bar{\lambda} \bar{x} \Rightarrow \bar{x}^T A^T = \bar{x}^T \bar{\lambda}$$

$$\Rightarrow \bar{x}^T A = \bar{x}^T \bar{\lambda}$$

$$\bar{x}^T A x = \bar{x}^T \bar{\lambda} x$$

$$\bar{x}^T A x = \lambda \bar{x}^T x$$

$$\bar{x}^T A \bar{x} = \bar{\lambda} \bar{x}^T \bar{x}$$

$$\bar{x}^T A x = \lambda \bar{x}^T x = \bar{x}^T A x = \bar{x}^T \bar{\lambda} x$$

$$\lambda \bar{x}^T x = \bar{\lambda} (\bar{x}^T x) \text{ if } \bar{x}^T x \neq 0$$

$$\lambda = \bar{\lambda} \quad (\lambda \text{ is Real})$$

length squared

$$\bar{x}^T x = [\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$\bar{x}_1 x_1 = (a+ib)(a-ib) = a^2 + b^2$$

$\bar{x}^T x$ is a positive number, of zero if x is zero

$$A = Q \Lambda Q^T \quad (A = A^T)$$

$$= [q_1 q_2 \dots q_n] \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix} \begin{bmatrix} q_1^T \\ q_2^T \\ \vdots \\ q_n^T \end{bmatrix}$$

Every symmetric matrix is a comb of perp - projection matrices

$$= \lambda_1 \underbrace{q_1 q_1^T}_{\uparrow \text{projection matrix}} + \lambda_2 q_2 q_2^T + \dots + \lambda_n q_n q_n^T$$

Signs of pivots are same as the signs of λ 's.

positive pivots = # positive λ 's

★ positive definite matrix
symmetric

$$\begin{bmatrix} 5 & 2 \\ 2 & 3 \end{bmatrix} \text{ pivots } 5, \frac{11}{5}$$

all eigen values are positive

all pivots are positive

all sub determinants: positive.

Linear Algebra · Lecture 26 Complex matrix; Fast fourier transformation. FFT

$$\begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ \vdots \\ z_n \end{bmatrix} \text{ in } \mathbb{C}^n$$

① length of z = $\bar{z}^T z$
 $z^H z$

H = Hermitian (conjugate + transpose)

② inner product: $y^H x$

③ Symmetric: $A^T = A$ in real, but not for complex.

Hermitian matrix $\bar{A}^T = A$
 $A^H = A$

eg. $\begin{bmatrix} 2 & 3+i \\ 3-i & 5 \end{bmatrix}$

unitary matrix

④ perpendicular:
 $q_1, q_2, \dots, q_n$

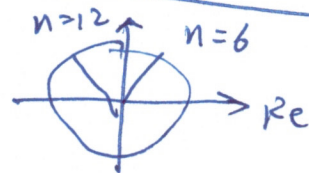
$$q_i^H q_j = \begin{cases} 0 & (i \neq j) \\ 1 & (i = j) \end{cases}$$

$$Q^H Q = I$$

$$F_n = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ w & w^2 & w^3 & \dots & w^{n-1} \\ w^2 & w^4 & w^6 & \dots & w^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ w^{n-1} & w^{2(n-1)} & w^{(n-1)^2} & \dots & w^{(n-1)^2} \end{bmatrix}$$

$$(F_n)_{ij} = w^{ij}$$

$$i, j = 0, \dots, n-1$$



$$w^n = 1 \quad w = e^{i2\pi/n}$$

$$n=4 \quad w^4 = 1 \quad w = e^{2\pi i/4} = i$$

$$= \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$$

$$F_4 = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & j^2 & j^3 \\ 1 & j^2 & j^4 & j^6 \\ 1 & j^3 & j^6 & j^9 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}$$

columns are orthonormal!

$$F_4^H F_4 = I.$$

$$[F_{64}] = \begin{bmatrix} I & D \\ I & -D \end{bmatrix} \begin{bmatrix} F_{32} & 0 \\ 0 & F_{32} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad D = \begin{bmatrix} w & & \\ & w^2 & \\ & & \ddots \\ & & & w^{31} \end{bmatrix}$$

Linear Algebra Lecture 27.

Positive Definite Matrix. (Tests)

Tests in Minimum ($x^T A x > 0$)

Ellipsoids in R^n

Test for positive definite:

- ① $\lambda_1 > 0 \quad \lambda_2 > 0$
- ② $a > 0 \quad ac - b^2 > 0$
- ③ pivots $a > 0$
 $\frac{ac - b^2}{a} > 0$
- ④ $x^T A x > 0$

Examples

pivot, 2

$$\begin{bmatrix} 2 & 6 \\ 6 & 18 \end{bmatrix}$$

Positive semi definite, $\lambda = 0, 20$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}^T \begin{bmatrix} 2 & 6 \\ 6 & 18 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \text{quadratic form}$$

$$= 2x_1^2 + 12x_1x_2 + 18x_2^2$$

$\begin{bmatrix} 2 & 6 \\ 6 & 7 \end{bmatrix}$ Graphs of $f(x,y) = \vec{x}^T A \vec{x}$



$$= ax^2 + 2bxy + cy^2$$

$$= 2x^2 + 12xy + 7y^2$$

↑
saddle

$$\begin{bmatrix} 2 & 6 \\ 6 & 20 \end{bmatrix} = \begin{bmatrix} 2 & 6 \\ 0 & 2 \end{bmatrix} \quad \leftarrow \text{pivots are coefficient of squares}$$

$$L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 6 \\ 6 & 20 \end{bmatrix}$$

$$\det = 4$$

$$\text{trace} = 22$$

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 2 & 6 \\ 6 & 20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 2x_1^2 + 12x_1x_2 + 20x_2^2$$

$$x^T A x > 0, \text{ except at } x = 0$$

Calculus: (single variable)

$$1^{\text{st}} \text{ Deriv} = 0$$

$$2^{\text{nd}} \text{ Min} \sim \frac{d^2 u}{dx^2} > 0$$

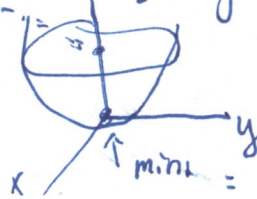
for Min

$$18.06. \text{ Min} \sim f(x_1, x_2, \dots)$$

Matrix of 2nd derivs is positive definite

$$z = 2(x+3y)^2 + 2y^2$$

$$f(x,y) = 2x^2 + 12xy + 20y^2$$



$$1^{\text{st}} \text{ derivs} = 0 \quad \left| \quad 2^{\text{nd}} \text{ derivs} > 0 \right.$$

Matrix of 2nd derivatives

($x^T A x \rightarrow A$ will be the 2n-d derivative matrix)

$$\begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} \text{ (Hessian)}$$

need to be PD to have minimum

3x3 example.

dets: 2, 3, 4

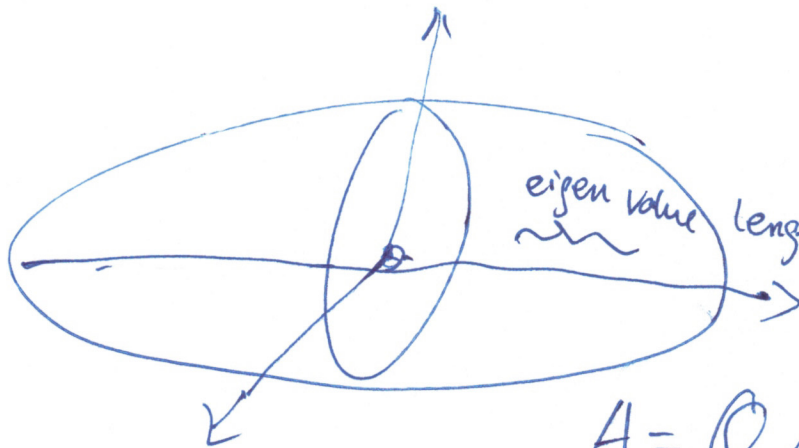
$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

pivots: $2, \frac{3}{2}, \frac{4}{3}$

eigenvalues: $2-\sqrt{2}, 2, 2+\sqrt{2}$

$$f: x^T A x = 2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_2 - 2x_2x_3 > 0$$

$f=1 \Rightarrow$ a ellipsoid in 3d.



$$A = Q \Lambda Q^T$$

Linear Algebra Lecture 28 :
 $A^T A$ is positive definite!!!

positive definite means:
 $x^T A x > 0$. (except for $x=0$)

If A, B are positive definite.
 $A+B$ is also positive definite.
 $x^T (A+B) x > 0$. ✓

Now A m by n , $\boxed{\text{rank} = n} \rightarrow$ (independent columns, no null space).

$A^T A$ square, symmetric, positive definite.

↓
 Like a square number!

$$x^T A^T A x = \underbrace{(Ax)^T (Ax)}_{\text{dot product of a vector, } \oplus} = \|Ax\|^2 \geq 0 \text{ (except } x=0)$$

Similar Matrix:
 $n \times n$ matrices

has the same λ !!

Same # of vectors

A and B are similar
 means for some M .

$$B = M^{-1} A M$$

Example: A is similar to Λ .

$\lambda = 3, 1$

$$S^{-1} A S = \Lambda \quad (\text{a family of matrix})$$

$$\begin{matrix} M^{-1} & A & M \\ \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix} & \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} & \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} \end{matrix} = \begin{bmatrix} -2 & -13 \\ 1 & 6 \end{bmatrix} \quad \begin{matrix} \Lambda & B \\ \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} & \begin{bmatrix} 3 & 7 \\ 0 & 1 \end{bmatrix} \end{matrix} \quad \begin{matrix} & & \\ & & \\ & & \end{matrix} \begin{matrix} \begin{bmatrix} 1 & 7 \\ 0 & 3 \end{bmatrix} \end{matrix}$$

Note = $A = S \Lambda S^{-1}$ if independent eigen vectors
 $A = Q \Lambda Q^T$ if A is symmetric
 $A = Q \Lambda^+ Q^T$ if Λ^+ is all positive pos def.

Similar matrices have the same eigen values!!! (same # of vectors)

proof:

$$Ax = \lambda x \quad (B = M^{-1}AM)$$

$$\Downarrow$$

$$AMM^{-1}x = \lambda x$$

$$\Downarrow$$

$$(M^{-1}AM)M^{-1}x = \lambda M^{-1}x$$

~~eigen vector~~

$$\Downarrow$$

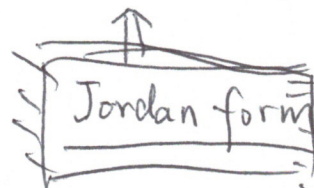
$$B(M^{-1}x) = \lambda(M^{-1}x)$$

$$\boxed{M^{-1}x \text{ is eigen vector}} \quad \boxed{\text{eigen value is } \lambda.}$$

BAD case: $\lambda_1 = \lambda_2 = 4$

one family has $\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \rightarrow M^{-1} \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} M = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$
small only

big family includes $\begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}$: only 1 eigen vector!



Not diagonalizable, otherwise it becomes $\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$.

more members of family

$$\begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}, \begin{bmatrix} 5 & 1 \\ -1 & 3 \end{bmatrix}, \begin{bmatrix} 4 & 0 \\ 1 & 4 \end{bmatrix}$$

$$\begin{bmatrix} a & * \\ * & 8-a \end{bmatrix} \quad \text{trace} = 8$$

$$\det = 16.$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \lambda = 0, 0, 0, 0.$$

2 vectors | 2 missing vectors
dim $N(A) = 2$.

not similar

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \lambda = 0, 0, 0, 0.$$

2 vectors
dim $N(A) = 2$ } not similar.

Jordan block. $\neq$ 1 vector per block

$$J_i = \begin{bmatrix} \lambda_i & 1 & & 0 \\ & \lambda_i & & 0 \\ & & \ddots & \\ 0 & & & \lambda_i & 1 \\ & & & & \ddots \end{bmatrix}$$

Every square A is similar to Jordan matrix J .

$$J = \begin{bmatrix} \boxed{J_1} & & \\ & \boxed{J_2} & \\ & & \ddots \\ & & & \boxed{J_n} \end{bmatrix}$$

blocks = # eigen values.

Good = n distinct Jordan blocks.

$J = \Lambda$, n Jordan blocks.

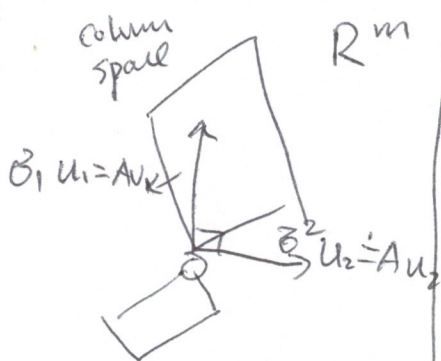
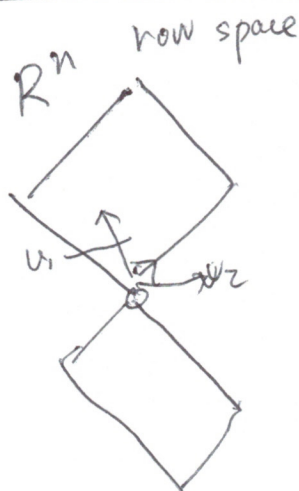
Linear Algebra 29. SVD = Singular Value decomposition.

$$A = U \Sigma V^T \quad \left| \begin{array}{l} \Sigma = \text{diagonal} \\ U, V, \text{ orthogonal} \end{array} \right.$$

symmetric definite positive.

$$A = Q \Lambda Q^T \leftarrow \text{special case for SVD}$$

$$A = S \Lambda S^{-1} \leftarrow \text{not SVD}$$



$$A \begin{bmatrix} v_1 & v_2 & \dots & v_n \end{bmatrix} \begin{array}{l} \text{Note} \\ r = \text{rank?} \\ v_{r+1}, \dots, v_n \\ \uparrow \\ \text{null space} \end{array}$$

$$= \begin{bmatrix} u_1 & u_2 & \dots & u_n \end{bmatrix} \begin{bmatrix} \sigma_1 & & 0 \\ & \sigma_2 & \\ 0 & & \ddots & 0 \end{bmatrix}$$

$$AV = U\Sigma$$

$$A = U\Sigma V^{-1}$$

$$= U\Sigma V^T$$

$$A^T A = V \Sigma^T U^T U \Sigma V^T$$

$$= V \begin{bmatrix} \sigma_1^2 & & 0 \\ & \sigma_2^2 & \\ 0 & & \ddots & 0 \end{bmatrix} V^T$$

$$A = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} \quad \begin{array}{l} v_1, v_2 \text{ in row space } \mathbb{R}^2 \\ u_1, u_2 \text{ in column space } \mathbb{R}^2 \\ \sigma_1 > 0 \quad \sigma_2 > 0 \end{array}$$

$$Av_1 = \sigma_1 u_1$$

$$Av_2 = \sigma_2 u_2$$

$$A^T A = \begin{bmatrix} 4 & -3 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} = \begin{bmatrix} 25 & 7 \\ 7 & 25 \end{bmatrix}$$

$\lambda_1 = 32, (1, 1) \cdot \frac{1}{\sqrt{2}}$
 $\lambda_2 = 18, (1, -1) \cdot \frac{1}{\sqrt{2}}$

$$\begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{32} & 0 \\ 0 & \sqrt{18} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$AA^T = U \Sigma V^T V \Sigma^T U^T$$

$$= U \Sigma^2 U^T$$

$$AA^T = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 32 & 0 \\ 0 & 18 \end{bmatrix}$$

$\lambda_1 = 32 \quad (1, 0)$
 $\lambda_2 = 18 \quad (0, 1)$

$$A^T A = Q \Lambda Q^T \quad v = Q$$

$$AA^T = Q' \Lambda Q'^T \quad u = Q'$$

Example 2.

$$A = \begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix}$$

$$v_1 = \begin{bmatrix} .8 \\ .6 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} .6 \\ -.8 \end{bmatrix}$$

row space

$$G \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

column space

$$c_2 \begin{bmatrix} 4 \\ 8 \end{bmatrix}$$

$$N(AT)$$

$$u_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \sqrt{125} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} .8 & .6 \\ .6 & -.8 \end{bmatrix}$$

A U V^T

$$A^T A = \begin{bmatrix} 4 & 8 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 4 & 8 \\ 8 & 6 \end{bmatrix} = \begin{bmatrix} 80 & 60 \\ 60 & 45 \end{bmatrix}$$

$$\lambda_1 = 0$$

$$\lambda_2 = 125$$

$v_1 \dots v_n$ orthonormal basis for row space

$u_1 \dots u_r$ orthonormal basis for column space

$v_{n+1} \dots v_n$ orthonormal basis for nullspace

$u_{r+1} \dots u_m$ for $N(AT)$.

and

$$A v_i = \sigma_i u_i$$

"choose the right space to connect these 4 spaces!!!"

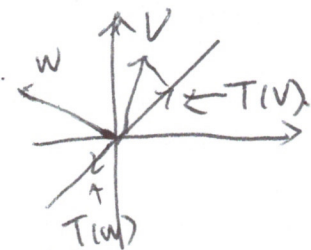
Connect the 4 spaces!!!

Linear Algebra Lecture 30.

Linear transformation.

Example 1. Projection

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$



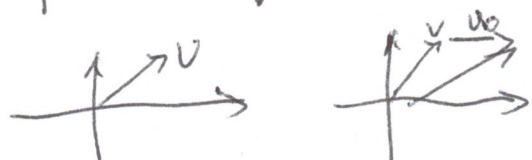
$$T(v+w) = T(v) + T(w)$$

$$T(cv) = cT(v)$$

$$T(cv+dw) = cT(v) + dT(w)$$

$$T(0) = 0$$

Example 2. Shift the whole plane.



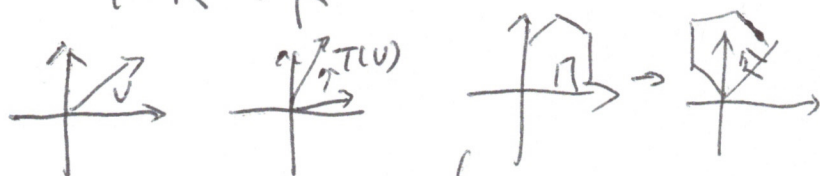
(not linear transformation!)

None example:

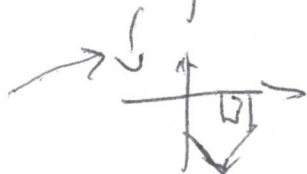
$$T(v) = \|v\|. \quad T: \mathbb{R}^3 \rightarrow \mathbb{R}^1 \quad T(-3v) = 9\|v\|$$

Example 2. Rotation by 45°

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$



$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



Example 3.!! Matrix A

$$T(v) = Av$$

$$A(v+w) = Av + Aw$$

$$A(cv) = cAv$$

Start: $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$

Example. $T(v) = Av \leftarrow A = \text{2 by 3 matrix}$

Information needed to know $T(v)$ for all inputs

$T(v_1), T(v_2), \dots, T(v_n)$ for any basis $v_1, \dots, v_n$

input.

$c_1, c_2, \dots, c_n$ are coordinates in basis

because $v = c_1v_1 + c_2v_2 + \dots + c_nv_n$

Coordinates come from a basis

$$v = c_1 v_1 + c_2 v_2 + \dots + c_n v_n$$

$$v = \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix} = 3 \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_{\text{standard basis}} + 2 \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{\text{standard basis}} + 4 \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{\text{standard basis}}$$

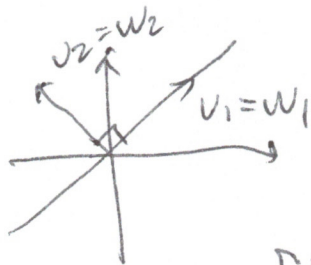
Construct matrix A that represents linear transformation T

$$T: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

Choose basis $v_1, \dots, v_n$ for input $\mathbb{R}^n$.

Choose basis $w_1, \dots, w_m$ for output in $\mathbb{R}^m$.

Want matrix A
Projection



$$v = c_1 v_1 + c_2 v_2, \quad (c_1, c_2)$$

$$T(v) = c_1 v_1, \quad (c_1, 0)$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} c_1 \\ 0 \end{bmatrix}$$

A = input
coords

output
coords.

(we choose the eigen vector basis)

leads to diagonal matrix $\wedge$

Proj onto 45° line. Use standard basis:

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = w_1, \quad v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = w_2$$

$$P = \frac{aa^T}{a^T a} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

Rule to find A . Given base $v_1 \dots v_n$
 $w_1 \dots w_n$

1st column of A : ~~Apply~~ $T(v_1) = a_{11}w_1 + a_{21}w_2 + \dots + a_{n1}w_n$
write

2nd column of A : $T(v_2) = a_{12}w_1 + a_{22}w_2 + \dots + a_{n2}w_n$

$$A \begin{pmatrix} \text{input} \\ \text{coords} \end{pmatrix} = \begin{pmatrix} \text{output} \\ \text{coords} \end{pmatrix}$$

$$T = \frac{d}{dx}$$

$$\text{Input} = c_1 + c_2x + \cancel{c_3x^2}$$

$$\text{basis} = 1, x, x^2$$

$$\text{output} = c_2 + 2c_3x$$

$$\text{basis} = 1, x$$

$$A \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} c_2 \\ 2c_3 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Lecture 31 = change of basis

Image compression. = 512×512 Pixel $[0 \sim 255]$

an image is a vector of $\mathbb{R}^{(512)^2}$

Standard basis

$$\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} \dots \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

Better basis

$$\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ -1 \end{bmatrix} \dots \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

Fouries basis 8×8

$$\begin{bmatrix} 1 \\ \vdots \\ w_1 \\ w_2 \\ \vdots \\ w_{n-1} \end{bmatrix}$$



JPEG

Signal
change basis
Coeffs c
compression
coeff $\hat{c}$ (may 0)

$$\hat{x} = \sum \hat{c}_i v_i$$

Standard basis

$$P = c_1 w_1 + c_2 w_2 \dots c_8 w_8$$

$$\begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_8 \end{bmatrix}$$

Wavelet basis

$$\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ -1 \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 0 \\ 0 \\ 0 \end{bmatrix} \dots$$

$$P = \begin{bmatrix} \text{Wavelet matrix} \\ W \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_8 \end{bmatrix}$$

$$= \text{solve } P = Wc$$

$$\hookrightarrow c = W^{-1}P$$

Good basis = ① Fast (FFT) (FWT)
② Few is enough.

Change of basis

columns of

W = new basis vectors.

$$\begin{bmatrix} x \end{bmatrix}_{\text{old basis}} \longrightarrow \begin{bmatrix} c \end{bmatrix}_{\text{new basis.}} \quad \boxed{x = Wc}$$

T with respect $v_1 \dots v_8$
it has matrix A
with respect $w_1 \dots w_8$
it has matrix B .

Similar: $B = M^{-1} A M$

What is A ? using basis $v_1 \dots v_8$.

Know T completely from $T(v_1), T(v_2) \dots T(v_8)$

because every x is a combination of basis.

$$x = c_1 v_1 + c_2 v_2 + \dots + c_8 v_8.$$

$$T(x) = c_1 T(v_1) + c_2 T(v_2) + \dots + c_8 T(v_8)$$

Write $T(v_1) = a_{11} v_1 + a_{21} v_2 + \dots + a_{81} v_8$ (coeff of col 1)

$T(v_2) = a_{12} v_1 + a_{22} v_2 + \dots + a_{82} v_8$ (coeff of col 2).

$$[A] = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{18} \\ a_{21} & a_{22} & \dots & a_{28} \\ \vdots & \vdots & \ddots & \vdots \\ a_{81} & a_{82} & \dots & a_{88} \end{bmatrix}$$

Eigen vector basis

$$T(v_i) = \lambda_i v_i$$

what is A .

$$A = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix} = \Lambda$$